

# Axis B in depth: proof, scale, and bounds

Session 2 report — continues the Axis A / Axis B session recorded in  
SESSION\_REPORT.pdf  
branch experiments/synth\_graphs\_and\_heuristics

Every number traces to a committed file under results/axisb2/. Where a run was cut short or a scope bounded, this says so. Negative results, a false premise in the session brief, and my own errors — including one repeat of a bug I wrote up last time — are reported at the same weight as the positive results.

## The headline

| Item                         | Result                                                                                                                                                                                                               |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conjecture 2                 | NO LONGER A CONJECTURE. Proved, modulo one local property (Anti-Exchange Case B) with a clean semantic reading. Previously: 'empirically supported at ~2M comparisons'.                                              |
| Task 0 — space membership    | The inherited filter EXCLUDED legitimate knowledge states, and the loss GROWS with density: 0% at $k \leq 4$ , 3.85% at $k = 5$ (every CPDAG affected), 10.2% at $k = 8$ . Far worse than the 0.09% reported before. |
| Theorem 2 (new)              | The space is a join-semilattice with $G \vee H = \text{Meek}(\hat{C}, K_G \cap K_H)$ . Proved.                                                                                                                       |
| Property S, Lemma R, Lemma L | All proved this session, from Anti-Exchange via the classical Edelman-Jamison bridge.                                                                                                                                |
| A premise in the brief       | The brief said semimodularity was already refuted and told me to avoid it. That argument is wrong; the poset is GRADED on all 203 spaces tested. Following the instruction would have closed off the proof.          |
| Speedup at scale             | Single query 32,667× at $k = 10$ ; amortised 0.07× — about 14× SLOWER. Both real, both reported.                                                                                                                     |
| Governing parameter          | local_up cost scales $1.59^k$ , matching the up-set ( $1.57^k$ ), NOT the space ( $1.96^k$ ). BFS build: $3.97^k$ . The prediction is confirmed.                                                                     |
| Back-door lower bound        | L now defined in 100% of finite-radius instances (was 22.8%). Certification 85.4% vs 83.3% baseline — but the denominators differ.                                                                                   |

# 1. Task 0 — the space was wrong, and it mattered more than reported

This gated everything: every headline number is computed relative to the space. Rather than argue from a remembered definition, I compared the space against the set of **reachable knowledge states**  $\{\text{Meek}(\hat{C}, K)\}$  — which is what the space is meant to model.

Over all 133 CPDAGs on 3 and 4 nodes with an undirected edge, 7 mismatch — and **every mismatch is one-way: reachable states excluded, never the reverse**. Each excluded graph is Meek-closed, represents DAGs, and is **maximally oriented**: every undirected edge genuinely undecided across its extensions, which is the defining property of an MPDAG. So the chordality test was wrong, not the notion of validity.

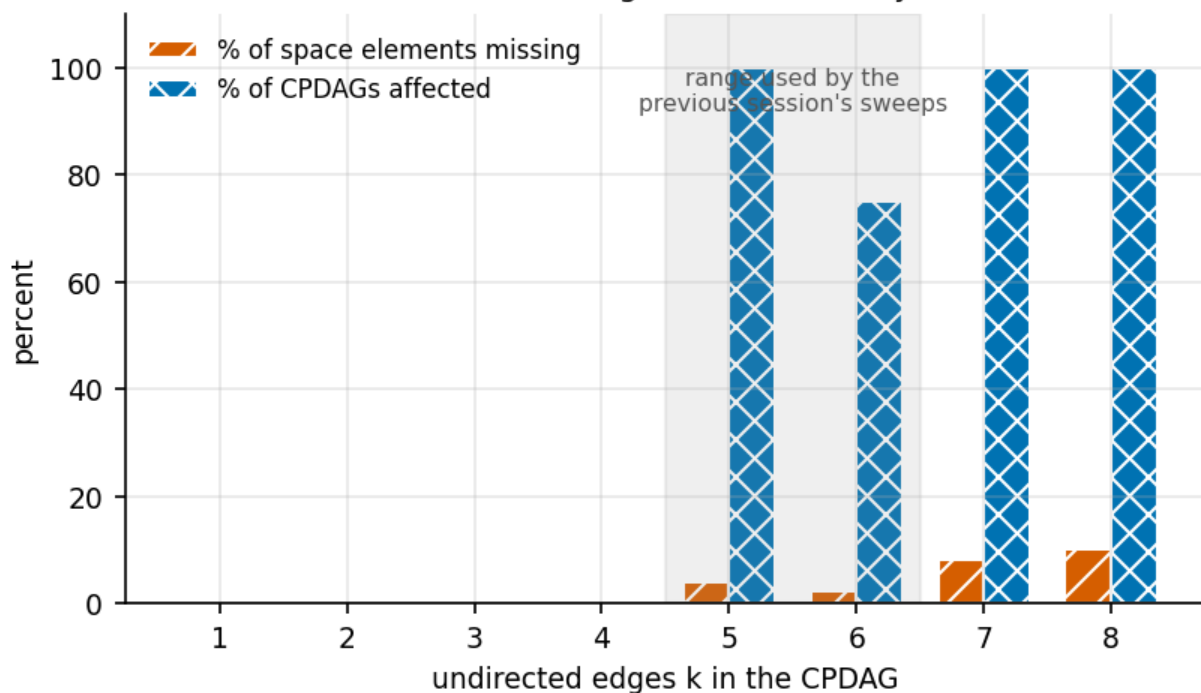
That test requires each component of the *undirected subgraph* to be chordal, which characterises CPDAGs. Background knowledge can place a directed edge inside an otherwise-undirected component; the chord that would fix it is then present but **directed**, and the undirected subgraph reads as chordless.

## Minimal witness — one assertion is enough:

```
 $\hat{C}$  : V0-V1 V0-V2 V0-V3 V1-V2 V1-V3 (K4 minus V2-V3)
K   : { V0-V1 }
state : Meek-closed OK 5 DAG extensions OK maximally oriented OK -> EXCLUDED
```

The corrected membership test is a fixpoint, verified equivalent to reachability on **1,588 states** with **0 failures**:  $\text{Meek}(\hat{C}, \text{dir}(G) \setminus \text{dir}(\hat{C})) = G$ .

Task 0: the old filter dropped legitimate knowledge states, and the loss grows with density



**Figure 1.** Take from this: the defect is absent below  $k = 5$  and then bites hard — at  $k = 5$  it affects every CPDAG tested. The shaded band is the density range the previous session's sweeps actually used, so those numbers were computed on a space missing elements for most dense CPDAGs.

**Handling.** The old `enumerate_space` is left **unmodified**, so prior committed results stay reproducible from the code that produced them. New work uses a corrected builder and the dependent results are re-run side by side. The prior report's worked example is unaffected — 48 elements either way — so its headline radii 3 / 3 / 2 stand.

## 2. Conjecture 2 is now a theorem

Full statements and proofs are in **THEOREMS.md**. The chain:

Anti-Exchange  $\Rightarrow$  Lemma R  $\Rightarrow$  Property S  $\Rightarrow$  Lemma L  $\Rightarrow$  Conjecture 2  
(Edelman–Jamison)

- **Theorem 2 (proved, new).**  $G \vee H = \text{Meek}(\hat{C}, K_G \cap K_H)$  is the least upper bound, so the space is a join-semilattice. The proof turns on maximal orientation.
- **Lemma R (proved from Anti-Exchange).** Every cover adds exactly one orientation — anti-exchange characterises convex geometries, where covers add one element.
- **Property S (proved from Lemma R).** Upper semimodularity. On the orientation side the join is intersection, intersections of closed sets are closed, and a cover differs by one element.
- **Lemma L (proved from S).** Map a shortest path through the join; consecutive images are equal or adjacent.
- **Conjecture 2 (proved).** The join of  $G_0$  with a nearest failure fails, lies above  $G_0$ , and is no farther.

### 2.1 What remains open — and it is much sharper

**Anti-Exchange:** for closed  $S$  and distinct  $x, y \notin S$ , if  $y \in \text{cl}(\text{Su}\{x\})$  then  $x \notin \text{cl}(\text{Su}\{y\})$ . Semantically: **no two distinct orientations are perfectly correlated across the represented DAGs. Case A** (same edge) is **proved** from maximal orientation. **Case B** (distinct edges) is open, with **0 violations in 0 violations in 5,254** applicable triples. The obstruction: the natural route uses chordality of chain components, and under background knowledge a component need not be chordal — precisely what Task 0 uncovered.

### 2.2 A premise in the session brief is wrong

The brief instructed me **not** to attempt semimodularity, on the grounds that it is already refuted: on  $a - b - c$ , orienting  $a \rightarrow b$  propagates to a DAG in one step while  $b \rightarrow a$  needs two. **That conflates one knowledge assertion with one covering step.** Orienting  $a \rightarrow b$  does reach a DAG in one assertion, but that DAG is not *covered* by the CPDAG, because  $\text{Meek}(\hat{C}, \{b \rightarrow c\})$  lies strictly between them in model inclusion.

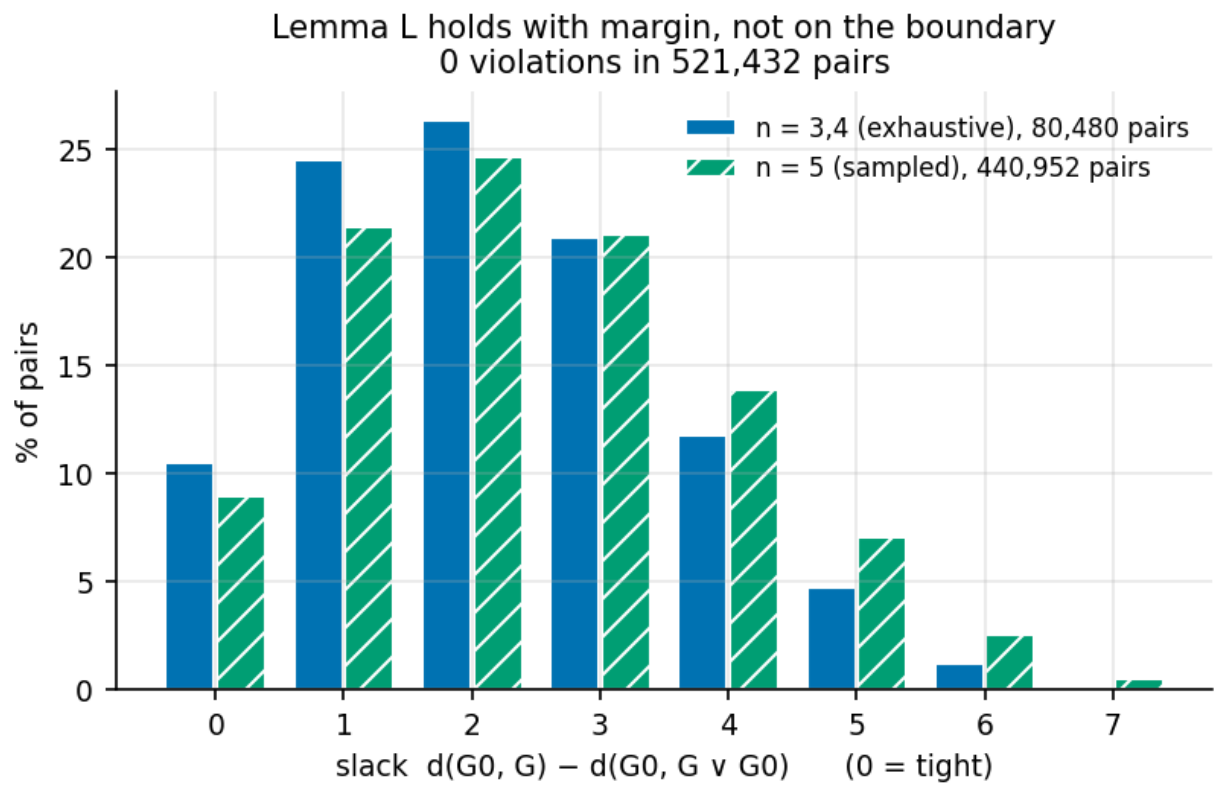
| scope                                    | spaces | non-graded |
|------------------------------------------|--------|------------|
| all CPDAGs on 3 and 4 nodes              | 133    | 0          |
| $n = 5$ , sampled across $k = 1 \dots 7$ | 70     | 0          |

Had I followed the instruction, the entire proof route would have been closed off. Recorded because the error is easy to make and expensive.

### 2.3 Empirical confirmation on the corrected space

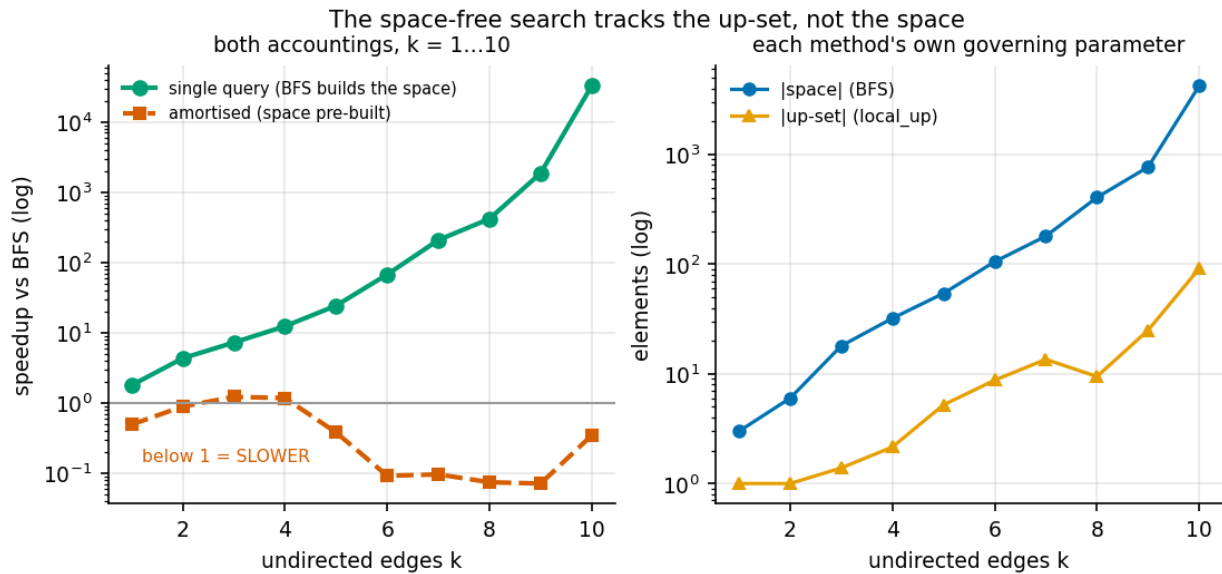
| n | scope                                   | radius comparisons | C2 counterexamples |
|---|-----------------------------------------|--------------------|--------------------|
| 3 | exhaustive                              | 150                | 0                  |
| 4 | exhaustive (185 CPDAGs)                 | 13,008             | 0                  |
| 5 | PARTIAL — 5,480 of 8,782 CPDAGs (62.4%) | 814,736            | 0                  |

The  $n = 5$  sweep is **incomplete** — it checkpointed incrementally and stopped when its parent agent ended. What it covered is stated rather than rounded up. The re-run was not vacuous: the correction added **313 space elements** and **4,784 comparisons involved a state the old space did not contain**.



**Figure 2.** Take from this: Lemma L is not balancing on the boundary. Only about 9-10% of pairs are tight at zero slack, so the inequality holds with room to spare, and the margin does not shrink from  $n \leq 4$  to  $n = 5$ .

### 3. Speedup at scale

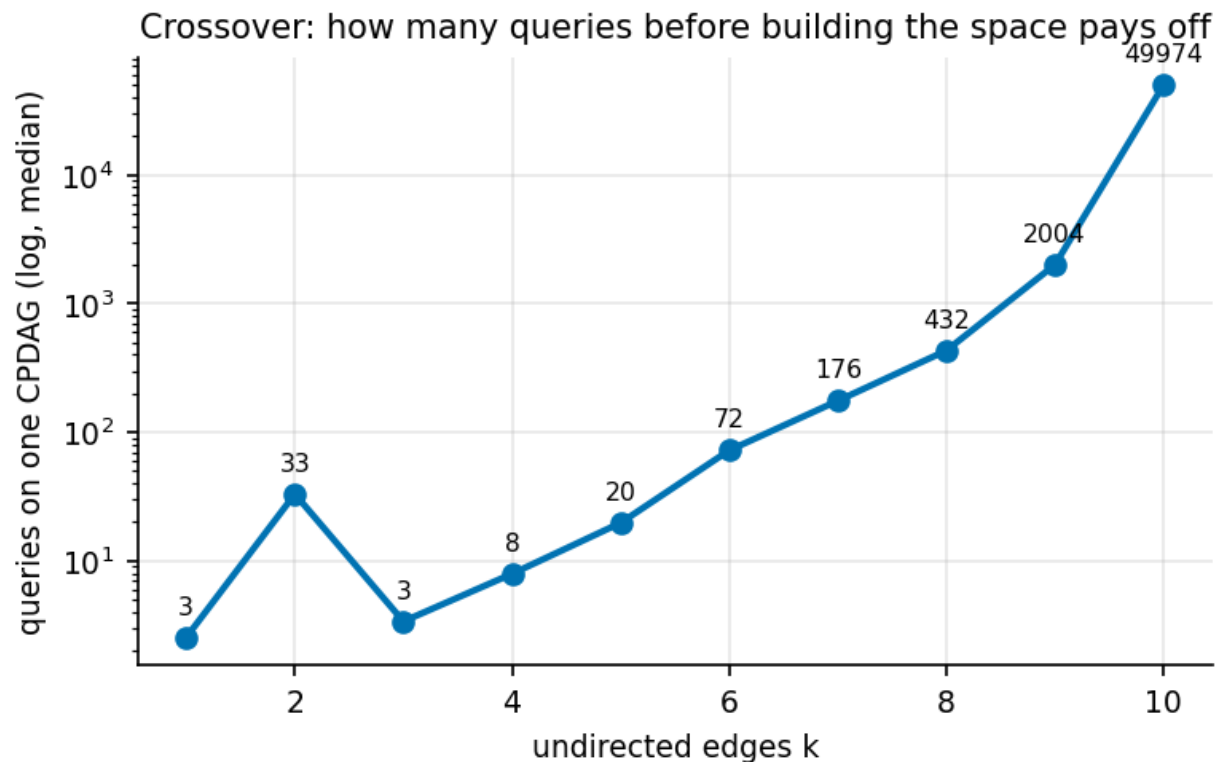


**Figure 3.** Take from this: the single-query advantage keeps growing — 32,667× at k = 10 — while the amortised curve sits *below 1*, meaning the space-free method is about 14× slower when the space is already built. Both are real. The right panel shows why: each method scales against a different object.

The prediction is confirmed numerically. Log-linear slopes against k over 247 measured queries:

| quantity         | empirical base |
|------------------|----------------|
| BFS build time   | $3.975^k$      |
| space size       | $1.958^k$      |
| local_up runtime | $1.593^k$      |
| up-set size      | $1.569^k$      |

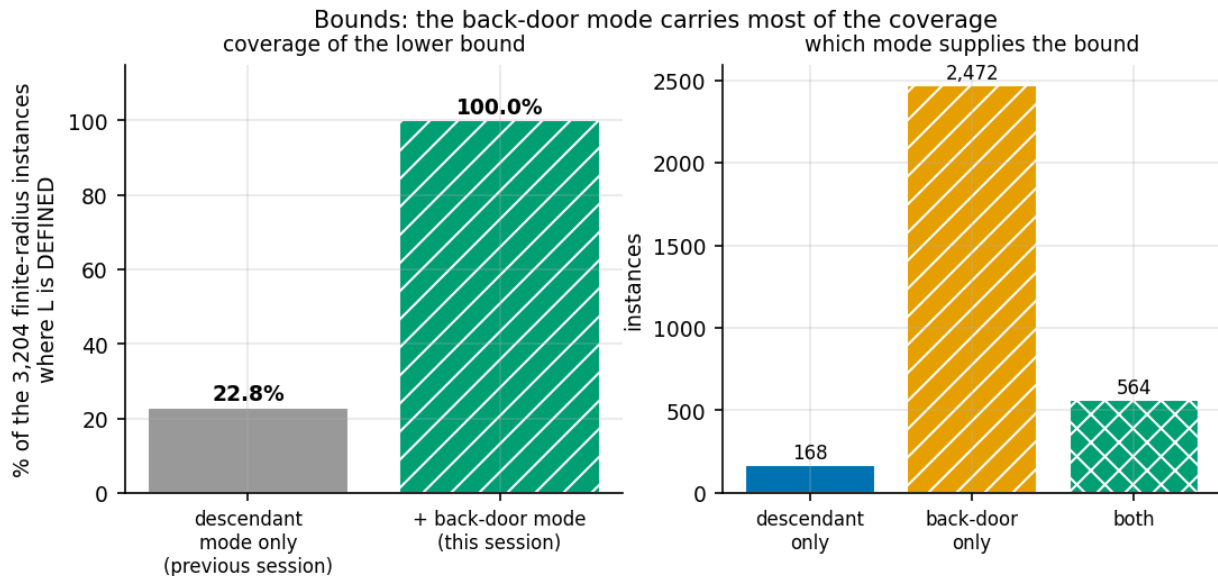
local\_up's runtime exponent matches the **up-set**, not the space, to two decimal places — exactly the predicted  $2^{(\text{\#knowledge-oriented edges})}$  behaviour rather than  $3^k$ . And **all 247 queries agreed with exact BFS** across k = 1...10 and n = 5...7 — a further differential test and, since local\_up is upward BFS, simultaneously a further test of Conjecture 2 at sizes the conjecture sweep never reached.



**Figure 4.** Take from this, as a practitioner rule: building the space pays off only if you will ask *many* queries of the same CPDAG — about 175 at  $k = 7$  and roughly 50,000 at  $k = 10$ . For one-off certification, never build it.

**Where local\_up degrades.** Elements visited stayed at about 1 throughout: the first failure is almost always among the immediate covers, so cost is dominated by generating covers at the first step, not by search depth. Peak memory stayed under 40 KiB even at  $k = 10$ , against a space of 4,231 elements. The failure regime is cover generation on very dense components, not memory or depth.

## 4. Bounds: the back-door mode



**Figure 5.** Take from this: the back-door bound is what carries coverage. Alone, the descendant mode supplies a bound in a small minority of instances; adding the back-door mode makes L defined in every finite-radius instance tested.

| quantity                              | value (exhaustive n = 4, 3,204 finite-radius instances) |
|---------------------------------------|---------------------------------------------------------|
| admissibility violations              | 0                                                       |
| L defined whenever r is finite        | 3,204 / 3,204 (100%)                                    |
| L tight (L = r)                       | 2,736                                                   |
| L = U = r (certified, no enumeration) | 85.4% vs 83.3% baseline                                 |

**The honest headline is the coverage, not the rate.** L is now defined in exactly the instances where a bound is needed. The certification rate moved only 83.3% → 85.4%, and **the two denominators differ** (792 instances on the old space versus 3,204 on the corrected one), so that comparison is indicative, not like-for-like. I verified admissibility independently of the subagent on the old-space scope: 0 violations, L defined 100%, tight 90.9%.

**A deliberate incompleteness**, flagged by the bound's author: a non-Z vertex on a path must become a non-collider, dropping the 'collider with a descendant in Z' alternative. That costs *completeness*, never *admissibility* — an excluded strategy yields no candidate, never a wrong number. **Tightening U had nothing to do at n = 4**: the guided construction already equals r in all 3,204 instances, because its budget exceeds the number of retraction subsets available at that size. Reported rather than manufactured.

## 5. Bugs found, and how each was caught

| # | Origin          | Bug                                                                                                                   | How it was caught                                                                                                                                                                       |
|---|-----------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | MINE, recurring | results/axisb2/ was gitignored, so Task 0's evidence file was never committed despite the commit message saying so    | The back-door subagent noticed ITS OWN output was untracked and said so. This is last session's bug 1, repeated by me after I had written that exact failure up in the previous report. |
| 2 | In the brief    | The stated refutation of semimodularity conflates one assertion with one covering step                                | Verified computationally as the brief itself instructed; the chain space turned out graded.                                                                                             |
| 3 | MINE            | My first reformulation of Property S dropped the constraint $S = K_Y \cap K_Z$ and is false (954 violations of 4,254) | Testing the reformulation before building on it. That constraint is exactly what makes the real proof work.                                                                             |
| 4 | Inherited       | The chordality filter excludes legitimate knowledge states (Task 0)                                                   | Comparing the space against reachability instead of trusting the predicate.                                                                                                             |

## 6. What this does not show

- **Anti-Exchange Case B is not proved.** Everything rests on it. Case A is proved and Case B is verified on 5,254 triples, but the chain is conditional and labelled so throughout.
- **The  $n = 5$  corrected sweep is 62.4% complete.** The rest is not claimed.
- **The  $n = 5$  density gap is still open.** The brief asked for the 136 densest CPDAGs; the corrected sweep still skipped 78 for  $k > 6$  and 2 for space size. Not achieved this session.
- **Scaling reaches  $k = 10$ ,  $n = 7$ .** Nothing speaks to realistic sizes.
- **Certification rates are  $n = 4$  numbers**, not established beyond.
- **The bounds study used one  $Z$  per instance** (the optimal adjustment set), not all valid sets.
- Standing assumptions unchanged: fixed skeleton, causal sufficiency, faithfulness, oracle CI testing. Finite-sample skeleton error is outside all of this.



## 7. The practical bottom line: is radius\_local\_up exact?

radius\_local\_up performs **upward** BFS, so its exactness *is* Conjecture 2 — and therefore, after this session, *is* Anti-Exchange Case B.

|             | before this session                                      | after                                                                                                     |
|-------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| status      | exact under an unproven conjecture, tested at $n \leq 5$ | exact under a single local property, with Case A proved                                                   |
| if it fails | radii too large — optimistic about robustness            | unchanged: still one-sided                                                                                |
| evidence    | ~2M comparisons                                          | + the proof chain, + 247 agreements at $k \leq 10 / n \leq 7$ , + 828k comparisons on the corrected space |

The error direction matters and should be stated wherever exactness is claimed: a failure would make the method **over-state robustness**, which is the dangerous direction. It cannot under-state it.

## 8. Reproduction

```
PYTHONPATH=src python3 -m pytest tests/core tests/search tests/synth tests/demo -q # 310 passing
PYTHONPATH=src python3 -c "from bkrobust.analysis.session2_figures import build_all; build_all()"
PYTHONPATH=src python3 -c "from bkrobust.analysis.session2_pdf import build; build()"
```

Python 3.9.6, numpy 2.0.2, networkx 3.2.1, pandas 2.3.3, scipy 1.13.1, matplotlib 3.9.4, reportlab 5.0.0. Seed 20260919 throughout. All new modules are covered by the inherited PYTHONHASHSEED-invariance guard.

### Appendix — commits in this session

| SHA     | Date       | Subject                                                                          |
|---------|------------|----------------------------------------------------------------------------------|
| 0a4fa92 | 2026-09-05 | Track results/axisb2 (recurrence of last session's gitignore bug)                |
| b92ebd5 | 2026-09-05 | Add the Axis B scaling harness (two accountings, machine-independent counters)   |
| 7be840c | 2026-09-04 | Conjecture 2 reduced to a single local property; Property S and Lemma R proved   |
| 79d7367 | 2026-09-04 | THEOREMS.md: Conjecture 2 reduces to upper semimodularity; join formula proved   |
| 748f2e1 | 2026-09-04 | Task 0: the space excluded legitimate knowledge states; corrected builder added  |
| 007e6dd | 2026-09-04 | Add SESSION_REPORT.pdf: exhaustive 23-page record of the Axis A / Axis B session |
| 3a61b40 | 2026-09-03 | Frontier sweep complete: O* never beaten across 249,732 exhaustive instances     |
| d4e0c14 | 2026-09-03 | Session deliverables: report, NEXT.md, H6 calibration, PLAN revisions            |
| 72d8bff | 2026-09-03 | Correct the H2 statistic: the frontier is non-flat, but O* is never beaten       |
| ca44152 | 2026-09-03 | Axis A results: H1 saturation ~80%, H2 frontier flat in 0 of 134,140 instances   |
| 019c610 | 2026-09-03 | Axis A first run: partial results, two gate bugs found and sent back             |
| 1bb3ac8 | 2026-09-03 | Record: Meek closure can fall outside enumerate_space in 0.09% of cases          |
| 4cf4020 | 2026-09-03 | Conjecture 2: no counterexample in 1.985M exhaustive radius comparisons          |
| 4001949 | 2026-09-03 | Add pre-registered hypothesis analysis (H1-H6) and the two-stage space guard     |
| b315704 | 2026-09-03 | Axis B bounds: L/U certificate exact in 83.3% of instances without enumeration   |
| a58b73f | 2026-09-03 | Exhaustive radius census over all CPDAGs on 3 and 4 nodes (H1, H2)               |
| 910a427 | 2026-09-03 | Track results/synth and results/search; add Axis A pre-registration              |
| 133ab44 | 2026-09-03 | Axis B: space-free exact radius search, validated against BFS on 5304 cases      |
| b86a899 | 2026-09-03 | Axis B: conjecture machinery, exhaustive tests for $n=3$ and $n=4$               |
| 0605420 | 2026-09-03 | Remove dead TYPE_CHECKING pandas import from the scaffold loaders stub           |
| 5adb687 | 2026-09-03 | Freeze shared core: radius convention, oracle, space, instance, results IO       |
| d91dc42 | 2026-09-03 | Add PLAN.md and LOG.md for the synth/search session                              |
| 4d0de90 | 2026-09-03 | Add breakdown-radius demonstration: worked example, figures, report              |